DATA AND REPORT – DUE ____ / ____ / ____ **SECTION _____**

* Show your work and keep track of significant figures. **DATE _____**

PARTNER _____

DATA

1. Number or letter on the bar: _____
2. Mass and appearance of the metal bar.
   a. Mass _____
   b. Shape _____
   c. Color _____
3. Dimensions of the bar:
   a. Longest side (Length) – or Diameter (for cylinders): _____
   b. Medium side (Height): _____
   c. Short side (Width): _____ (not needed for cylinders)
4. Water displacement of the bar:
   a. Initial Water Volume _____
   b. Final Water Volume _____
5. Mass and appearance of the aluminum shot:
   a. Mass _____
   b. Shape _____
   c. Color _____
6. Water displacement of the aluminum shot:
   a. Initial Water Volume _____
   b. Final Water Volume _____

RESULTS

* Show your work and keep track of significant figures.

7. Calculate the metal bar's volume (in cm^3) using its dimensions.

8. Calculate the volume of the bar in mL using water displacement.

9. Chemists compare two values mathematically by calculating percent difference. Calculate the percent difference between the two volumes using the following equation:

$$\%\ Difference = \frac{Volume\ from\ dimensions - Volume\ by\ displacement}{Volume\ by\ displacement} \times 100$$

10. How does the bar's 'volume measured by dimensions' compare to the bar's 'volume measured by displacement'? (Use complete sentences, mentioning both numerical volume measurements, and the percent difference)

11. Calculate the density of the metal bar. (Use the volume from the displacement measurement.)

12. Determine the density of the aluminum shot.

13. Compare the densities of the aluminum shot to the density of the metal bar you measured earlier. Could your bar be made of aluminum? Why or why not? If the bar is not aluminum, what could the metal be?

Metal	**Density (g/cm^3)**
Brass	8.5
Cast Iron	6.7
Copper	9.0
Gold	19.3
Lead	11.3
Silver	10.5
Stainless Steel	7.5
Steel	8.0
Titanium	4.5
Tin	5.8